

Supplementary materials for
Greening while Governing the Economy: Do Green
Central Banks Matter?

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1 Materials

1.1 Dependent Variable: Greenhouse Gas Emissions

My dependent variable consists of greenhouse gas (GHG) emissions. The data have been collected from the Emissions Database for Global Atmospheric Research (EDGAR) developed by the European Commission’s Joint Research Centre (1). The dataset is publicly available (https://edgar.jrc.ec.europa.eu/dataset_ghg2025) and has been widely used (e.g., 2, 3). EDGAR has separate estimates for countries’ total carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and fluorinated gases, as well as total GHG emissions measured in millions of CO₂ equivalent, which is the one I use in this article. From the original spreadsheets downloaded from EDGAR, I extracted the total GHG emissions data for the years 2001 to 2021.

EDGAR details its methodology to calculate emissions here: <https://edgar.jrc.ec.europa.eu/methodology>. Specifically, EDGAR quantifies global anthropogenic emissions by applying a consistent estimation framework across all countries and economic sectors by year. This approach ensures a highly comparable data baseline across heterogeneous regulatory regimes and distinct gas categories. Only two countries needed to be excluded from the assessment due to missing data on the dependent variable: Marshall Islands and Northern Mariana Islands.

I use GHG emissions as a dependent variable based on three specifications. First, given the skewed distribution of GHG emissions across countries and years, I transformed this variable to a logarithmic format. Additionally, I normalized GHG emissions by the log of countries’ total population to obtain a measure of emissions per capita. I also normalized GHG emissions by the log of gross domestic product (GDP) per capita (PC) to account for economic and consumption differences across countries. The normalization process involves taking the logarithm of the ratio of GHG emissions to the population or GDP PC. An example of emissions per capita is expressed mathematically as:

$$\log(ghgpc_{i,t}) = \log(ghg_{i,t}) - \log(pop_{i,t}) \quad (1)$$

where $\log(ghgpc_{i,t})$ is the log-transformed emissions per capita variable for country i in year t . $\log(ghg_{i,t})$ is the log of GHG emissions. $\log(pop_{i,t})$ is the normalization variable in its log-transformed version. In the Equation 1 example, I use population, but the same process applies to GDP per capita. I explain from which repositories the other variables have been collected and how I have built a merged dataset below.

1.2 Independent and Control Variables

1.2.1 NGFS

My main independent variable is membership in the Network for Greening the Financial System (NGFS), which serves as a treatment indicator of financial regulators’ commitment to incorporate climate risks into financial supervision. The NGFS maintains their membership directory updated and publicly available here: <https://www.ngfs.net/en/about-us/members>

[hip](#). However, the NGFS does not specify the year in which financial supervisors have become members or observers. In March 2024, NGFS officers provided a spreadsheet containing the years in which each member and observer joined the network until then. Since this is not available through the website above, the e-mail exchange and the original spreadsheet shared by the NGFS are available upon request for reviewers and editors. I excluded observers and restricted the NGFS data to members only between 2001 and 2021.

Subsequently, I identified each country where financial supervisors are located and recorded the year a nation joined the NGFS. This way, countries are not double counted. Abu Dhabi, Dubai, and the European Union (EU) were also removed from the data. Abu Dhabi and Dubai count as “United Arab Emirates”, whereas several EU nations appear independently in the dataset. I then created a dummy variable, **ngfs**, indicating NGFS membership and assigned a value of 1 to countries in the year in which they became NGFS members and 0 to non-members. Thus, non-members include nations *before* they joined the NGFS and countries that never became members within the period analyzed. Consider, for example, how France has been coded in Table [S1](#), as the Banque de France helped found the NGFS in 2017 (4).

Table S1: Example of NGFS Membership Coding: France (2015–2021)

Country	Year	NGFS Member
France	2001	0
France	2002	0
France	2003	0
France	2004	0
France	2005	0
France	2006	0
France	2007	0
France	2008	0
France	2009	0
France	2010	0
France	2011	0
France	2012	0
France	2013	0
France	2014	0
France	2015	0
France	2016	0
France	2017	1
France	2018	1
France	2019	1
France	2020	1
France	2021	1

As a result of the steps above, I built an initial *country* $\times$ *year* dataset, which I merged with other repositories from which other variables were obtained. In sum, the cross-sectional unit i is a country, which appears in each year t over a period of time. I detail other methodological decisions that resulted in the tidy, merged, final dataset used for the analysis in Section 2.

1.2.2 Central Bank Independence Index

A second independent variable of interest is central bank independence (CBI) (5, 6), which is publicly available (<https://sites.google.com/site/carogarriga/cbi-data-1>). CBI is an index ranging from 0 to 1, where 0 represents no independence and 1 denotes maximum independence (5). The index considers four formal attributes of central banks: (i) personnel independence of a central bank’s executive leadership; (ii) having exclusive power or not over price stability mandate in terms of policy objectives; (iii) independence in policy formulation to devise rules, norms, and regulations; and (iv) lending limitations when furnishing money to the own government (5). These metrics are compiled into two operational variables: an unweighted index that computes the simple average of all four components; and a weighted

index that mathematically penalizes loose sovereign lending rules by weighting restriction variables more heavily.

I use Garriga’s (5) weighted CBI index for two reasons. First, it “represents a 46 percent extension of the coverage of the most comprehensive dataset on de jure CBI available to date (Garriga 2016). On average, this dataset includes fifty four more countries per year than Bodea and Hicks (2015b, updated to 2020), thirty seven more than Romelli (2022), and twenty two more than Garriga (2016)” (5). And second, it offers more conservative estimates than other CBI datasets: “Due to systematically missing data on countries for whom information is harder to obtain, previous research may have overestimated the level of CBI in the world, which may have biased inferences based on smaller samples” (5). Therefore, this dataset provides more data on developing nations over a longer period of time than other repositories, and it helps address potential issues with overestimating CBI. As a robustness check, I also tested the unweighted index and found similar results (see Section 3).

1.2.3 World Bank

Following Chesler et al. (7), a set of control variables includes data from the World Bank, which have been collected using the `wbstats` package in R (8). To normalize the distribution of the controls either as a logarithm or square root, they were transformed based on the closest to a normal distribution displayed using the `gladder` function of the `describedata` package also in R (9). Essentially, `gladder` plots a series of potential variable transformations while showing a normal distribution line on each transformation. The indicators are the following.

Log of GDP per capita accounts for a country’s standard of living and its population purchasing power, which has long been positively correlated with higher emissions (7). **Log of GDP per capita growth**, in turn, measures economic growth and whether the income level per person has increased or decreased, and it also reflects an increase in emissions (7). **Log of population growth** follows demographic trends regarding population size, which yields inconclusive results for emissions when controlling for other factors like GDP per capita.¹ **Log of total natural resources rented** is likely positively associated with higher emissions (7), and it consists of the sum of, e.g., oil, forests, and mineral rents, among other assets, rented as a percentage of a nation’s GDP. **Log of trade dependency** is measured as the ratio of a country’s total trade to GDP, and the circulation of products tends to increase emissions (7). **Square root of agricultural land** refers to arable areas used either for temporary or permanent crops as a percentage of total land, which may also produce higher emission levels (7). And, conversely, the **square root of forest land** refers to areas natural or planted trees as a percentage of total land, which might help offset emissions increased by land use and land cover change.

¹This has been a long debate that traces back to Ostrom’s (10) work on challenging the assumption that population growth inherently leads to the depletion of natural resources.

1.2.4 Varieties of Democracy

Also drawing from Chesler et al. (7), as well as Kakenmaster (2), political indicators were collected from the Varieties of Democracy (VDM) project using the `vdm` package in R (11, 12). The VDEM variables are the following. **Political corruption index** (`v2x_corr`) ranges from 0 to 1, corresponding to the less corrupt to the more corrupt nations. Corruption levels shape the relationship between political regimes and GHG emissions (2, 13), with growing levels of corruption being associated with higher emissions. **Deliberative democracy index** (`v2x_delibdem`) also ranges from 0 to 1, with higher values denoting more effective deliberative democratic processes and a focus on common goods. As deliberations are key in climate and environmental governance (10, 14), it is expected a negative relationship between deliberative democracy values and GHG emissions. Finally, following Vaccaro (15), I select a variable of **state capacity** (`v2clrspct`) that captures the extent to which state agents can abide to the law while pursuing their mandates while arbitrariness and biased public administration are limited. I employed min-max re-scaling for state capacity to range from 0 to 1,² such as the other VDEM indicators. The higher the score, the higher the levels of capacity, which, in turn, are presumed to be associated with the ability to integrate financial and climate policies.

2 Methodology

Methodological decisions often entail considering trade-offs, particularly with respect to dropping observations and selecting a timeframe for analysis. Here, the example displayed in Table S1 raises two main issues. First, by creating a dataset spanning 2001 and 2024, the pre-treatment period—before 2017—has a disproportionate number of years for which a country being an NGFS member equals 0. This issue could overestimate the magnitude of the treatment effect.

Second, several control variables obtained from the World Bank had missing data for the years between 2021 and 2024, when I received the NGFS membership data. Because this is one of the first studies relying on NGFS participation as a treatment indicator using real-world data with observed values, I have prioritized to be as conservative as possible with missing data. Hence, I use listwise deletion. As Pepinsky (16) notes, “Only knowing the true values from the fully observed data allows us to compare the performance of [listwise deletion and multiple imputation]. But this exercise clearly reveals that multiple imputation can change substantive conclusions in real-world applications, and in this case, those conclusions will frequently be more misleading than those that come from listwise deletion.” I acknowledge that this approach might present issues also discussed by Pepinsky (16). However, since I use indicators based on real-world observed values and I aim to estimate conservative coefficients

²The min-max normalization is mathematically expressed as

$$\text{State Capacity}_{i,t}^{\text{scaled}} = \frac{\text{State Capacity}_{i,t} - \min(\text{State Capacity})}{\max(\text{State Capacity}) - \min(\text{State Capacity})}$$

where $\text{State Capacity}_{i,t}$ represents the raw state capacity score for country i in year t , and $\min(\text{State Capacity})$ and $\max(\text{State Capacity})$ are the sample minimum and maximum values, respectively.

employing causal inference techniques, my main goal has been to work using a balanced panel dataset.

To deal with the first issue above, I further restricted the initial dataset to the period between 2015—almost immediately before the establishment of the NGFS in 2017—and 2021. And, to address the second issue, I ensure that each country appears in every single year with no missing data on the dependent and independent variables. These steps result in a balanced panel dataset that is suitable for both standard TWFE modeling, as well as subsequent and more robust difference-in-differences analysis. The DID with TWFE models are based on a *country* $\times$ *year* sample of 133 nations that appear over 7 years, resulting in $n = 931$.

2.1 Descriptive Statistics

For an overview of the data, I break down the descriptive statistics by NGFS membership. Table S2 highlights the differences between groups. This includes nations before they received the treatment, after the treatment, and countries that never joined the NGFS.

Table S2: Descriptive statistics by NGFS membership treatment (N = 931)

Variable	Member N = 193	Non-Member N = 738	Differ- ences p-value
Greenhouse gas emissions	11.83 (1.82)	10.83 (1.74)	<0.001
Greenhouse gas emissions per GDP per capita	1.97 (2.16)	2.40 (1.85)	0.004
Greenhouse gas emissions per capita	-4.87 (0.59)	-5.44 (0.95)	<0.001
Trade (% of GDP)	4.41 (0.59)	4.25 (0.49)	0.005
GDP per capita (constant 2015 US\$)	9.86 (1.07)	8.43 (1.37)	<0.001
GDP per capita growth (annual %)	3.93 (0.11)	3.94 (0.10)	0.5
Natural resources	-0.86 (2.46)	0.60 (2.06)	<0.001
Population growth (annual %)	2.52 (0.07)	2.58 (0.14)	<0.001
Agricultural land (% of land area)	5.83 (2.00)	6.19 (1.82)	0.082
Forest area (% of land area)	5.54 (1.83)	4.93 (2.33)	0.003
Deliberative democracy index (0-1)	0.60 (0.24)	0.40 (0.23)	<0.001
Central Bank Independence Index (Weighted)	0.66 (0.20)	0.62 (0.19)	<0.001
Region			
East and South Asia and Pacific	40 (21%)	107 (14%)	
Europe and Central Asia	95 (49%)	213 (29%)	
Latin America and Caribbean	23 (12%)	103 (14%)	
Middle East and Africa	29 (15%)	307 (42%)	
North America	6 (3.1%)	8 (1.1%)	

The values correspond to the mean for continuous variables and the proportion for categorical variables. The standard deviation for the continuous variables is shown in parentheses.

Table S2 reveals some significant differences worth noting. NGFS members ($n = 193$) emit substantially more GHG in absolute terms than non-members ($n = 738$) (mean log GHG = 11.83 vs. 10.83, $p < 0.001$). This gap reflects the NGFS structure: Early adopters are predominantly industrialized economies whose central banks have the capacity and political motivation to engage with international financial governance (4). Compared to non-members, NGFS members are notably wealthier (log GDP per capita: 9.86 vs. 8.43, $p < 0.001$), more trade-exposed (4.41 vs. 4.25, $p = 0.005$), and score higher on deliberative democracy (0.60 vs. 0.40, $p < 0.001$). At the same time, emissions per GDP per capita are lower among NGFS members as opposed to non-members (1.97 vs. 2.40, $p = 0.004$). These systematic differences underscore that standard before-after or cross-sectional models may produce unreliable estimates of the NGFS effect on emissions, motivating the within-country identification strategy I employ here.

Indeed, the regional distribution of NGFS membership reinforces this concern. Europe and Central Asia account for nearly half of all member-country observations (49%) but less than a third of non-member observations (29%). By contrast, Middle East and Africa, which represent 42% of non-member observations, represent only 15% of NGFS member observations. Yet, GDP per capita growth does not differ significantly between members and non-members (3.93 vs. 3.94, $p = 0.5$). This similarity in growth trajectories matters: It suggests that the emissions reduction are not explained by members simply growing more slowly than non-members. Instead, it is consistent with pathways to decarbonization rather than economic deceleration. Relatedly, central bank independence is slightly but significantly higher among members (0.66 vs. 0.62, $p < 0.001$). This pattern suggests that central banks need some level of statutory insulation from short-term political interference before they can extend their monetary mandate to design policies targeted at handling environmental externalities.

The differences in Table S2 have both statistical and practical implications. Substantively, NGFS membership has thus far been concentrated among nations whose central banks already operate in well-developed regulatory environments. This feature may amplify the conditional effects related to the independence of financial supervisors. Empirically, the variation in CBI among NGFS members and non-members is modest in magnitude but important to further test whether and how CBI works as a moderator or mediation variable in the models discussed in Section 2.2. Finally, the sample differences raise questions about whether the findings generalize beyond the current NGFS membership composition, which I address by conducting robustness checks (see Section 3).

i Note

I need to double check how best to conduct the robustness check above on sample composition.

2.2 Regression Results

3 Robustness Checks

4 Replication Notes

The documents and code to replicate the results are available upon request, and they will be published as part of a GitHub public repository.

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